

1. **Bus Fare Minimization Problem** In a city, a bus stops every kilometer. Also, the bus fare is different from 1km to 10km, so you can use each section well to decrease the total fare to get to the destination. For example, let's suppose a bus fare list as below: 12 21 31 40 49 58 69 79 90 101 According to the list, the minimum fare would be 147 for 15km: → 1 ticket for 3km + 2 tickets for 6km. When the bus fare and distance are given, calculate the minimum fare that you can move the given distance. Input: - First line: 10 integers, the bus fare from 1km to 10km. - Second line: The distance to move, N. ($1 \leq N \leq 10000$) Output: - The minimum fare to travel the distance. Example: Input: 12 21 31 40 49 58 69 79 90 101 15 Output: 147

2. **Farmer's Shortest Path Problem** The farmer wants to return home, Farm No. 1, from Farm No. N. The farmer is very tired and wants to return home as fast as possible. In the farms he cultivates, there are T two-way roads. Create a program that helps the tired farmer find the shortest distance from Farm No. N to Farm No. 1. Input: - First line: T N M - T = number of roads ($1 \leq T \leq 10000$) - N = number of farms ($1 \leq N \leq 1000$) - M = number of roads (length ≤ 100) - Next T lines: S E D - Road between farm S and E with length D. Output: - The shortest distance from Farm No. N to Farm No. 1. Example: Input: 5 5 1 2 20 2 3 30 3 4 50 4 5 70 1 5 200 Output: 120

3. **Negative Temperature Storage Problem** A cold storage company has N storage units for various products. The company has received N orders that must be preserved at negative temperatures. The manager wants to identify which products must be preserved at negative temperatures. Input: - First line: an integer numOfProducts ($0 \leq \text{numOfProducts} \leq 1e6$) - Second line: N integers (temperature[i]), where $-1e6 \leq \text{temperature}[i] \leq 1e6$ Output: - The number of products that must be preserved at negative temperatures. Example: Input: 7 9 -3 8 -6 7 8 10 Output: 2

4. **Largest Plot of Land Problem** In a city there are N houses. A buyer is looking for a plot of land in the city on which to build their house. The buyer wants to find a plot of land between two houses where they can build the largest possible house. All houses in the city lie in a straight line, and their position is given relative to the entry point. Input: - First line: two integers num val (number of houses, always equal to val). - Next N lines: two integers per line H_i P_i, where H_i = house number, P_i = position of house. Output: - Two house numbers (in ascending order) between which the largest plot of land is available. Example: Input: 4 4 1 2 2 4 3 7 4 11 Output: 3 4